

Project – Sweet Market: Production Planning and Pricing Competition

INDE 302 Project

Due Date: November 20, 2025

Introduction

In this project, each team represents a small **artisan cake shop** that produces and sells ten distinct types of cakes. The goal of each team is to strategically plan its operations—deciding how much to invest in production capacity, how to price each product, and how to allocate resources—in order to achieve the **highest profit** among all competing teams.

The business environment consists of three distinct sales channels:

- **Local Shop:** Direct sales to walk-in customers. Demand is moderate but prices can be kept relatively high due to personalized service and premium experience.
- **Supermarket:** Bulk orders at discounted prices. Prices are generally lower, but the volume of demand can be considerably higher.
- **Online Local:** Delivery-based sales through online platforms. Prices fall between those of the local shop and the supermarket, but there are additional delivery and service fees.

Each channel behaves differently: the demand patterns, transportation costs, and service capacities vary across them. Accordingly, **the optimal prices and production allocations should differ per channel**. Students must make pricing decisions strategically, balancing margin versus volume.

The production of cakes involves multiple stages, each consuming specific resources:

- **Preparation:** Mixing ingredients and setting up workstations. Labor-intensive and charged according to hourly wages.
- **Baking:** Conducted in batches using an oven. The oven has a fixed batch size, and a baking time per batch. Electricity costs accrue for each minute of oven usage.
- **Decoration:** Requires skilled labor. Costs depend on the time per cake and the hourly wage.
- **Packaging:** Includes material cost and labor time for each cake. Each unit also incurs a fixed packaging cost.

All necessary data—including ingredient quantities, labor requirements, oven capacities, and energy costs—are provided in the associated data files. The total weekly cost of production will therefore depend on ingredient consumption, labor hours, oven electricity, and packaging expenses.

Files Provided

The following datasets are provided to each team:

File	Description
<code>instructor_demand_competition.csv</code>	Contains the parameters α_{ij} and β_{ij} for each cake–channel pair, which determine the relationship between price and demand.
<code>bills_of_materials.csv</code>	Lists the ingredients required for each cake, specifying the amount of each ingredient per unit of cake.
<code>cakes.csv</code>	Describes the production process for each cake, including batch size, oven time per batch, preparation time per unit, packaging time per unit, packaging cost per unit, and minimum production quantities.
<code>channels.csv</code>	Lists the market channels (Local, Supermarket, Online Local) with corresponding transportation cost per unit and service capacity.
<code>ingredients.csv</code>	Provides ingredient prices and units of measurement.
<code>wages_energy.csv</code>	Includes labor wage rates, oven rental rate, and energy costs.

Stage 1: Initial Investment

Each team begins with an initial capital of **\$4,000** and no resources. This amount must be allocated among the following investment categories:

- **Raw materials:** Purchase of ingredients to be used during production.

- **Labor:** Investment in preparation and decoration workforce capacity (weekly labor hours).
- **Oven capacity:** Purchase or rental of oven time (baking capacity measured in minutes per week).

Your chosen investment determines your available capacity for preparation, baking, and packaging operations. Teams cannot exceed the total available investment budget.

Stage 2: Pricing and Demand Estimation

Each team must determine the selling price P_{ij} of every cake i in each channel j . The weekly demand for each product–channel pair is computed according to the following deterministic equation:

$$D_{ij} = \alpha_{ij} - \beta_{ij}P_{ij},$$

where

- D_{ij} = weekly demand for cake i in channel j ,
- α_{ij}, β_{ij} = demand parameters (provided),
- P_{ij} = price of cake i in channel j .

The resulting demand will differ across channels. Supermarkets will typically require lower prices to stimulate volume, while local-shop customers may tolerate higher prices. Each team must therefore find a profitable balance between high margins and large quantities.

Stage 3: ILP Formulation for Optimal Production Planning

Using your investment decisions and demand forecasts, formulate an **Integer Linear Programming (ILP)** model to determine the optimal weekly production plan.

Decision Variables

- P_{ij} : price of cake of type i sold through channel j .
- y_{ij} : number of cakes of type i produced and delivered to channel j .
- s_{ij} : number of cakes of type i sold in channel j .
- b_i : number of produced batches of cake type i .

Objective Function

The objective is to maximize total profit:

$$\max \Pi = \sum_{i,j} P_{ij} \cdot s_{ij} - C_{\text{ingredients}} - C_{\text{labor}} - C_{\text{utilities}} - C_{\text{transportation}},$$

where:

- $C_{\text{ingredients}}$: cost of all ingredients consumed (from bills of materials and ingredient costs),
- C_{labor} : total cost of preparation, baking, decoration, and packaging times multiplied by wage rate,
- $C_{\text{utilities}}$: electricity and oven usage cost,
- $C_{\text{transportation}}$: transportation cost.

It is clear that the objective function contains products of decision variables ($P_{ij} \cdot s_{ij}$), which makes it non-linear. One way to restore linearity is to treat P_{ij} as inputs, instead of decision variables. You can experiment with different values of P_{ij} and find the associated optimal production plan.

Only the minimum between production quantity and demand is sold, ensuring that:

$$s_{ij} = \min(D_{ij}, y_{ij}).$$

The expression $s_{ij} = \min(D_{ij}, y_{ij})$ cannot be written directly in a linear program. To model it properly, use a set of *linear inequalities* that ensure it behaves like the minimum of the two quantities. Remember that if you are *maximizing* profit and s_{ij} appears with a positive coefficient in the objective, it will automatically take the largest feasible value allowed by those inequalities.

Constraints

- **Capacity constraints:** total time used in each production stage (preparation, baking, packaging) cannot exceed available capacities determined by investment.
- **Ingredient constraints:** total ingredient usage cannot exceed available stock.
- **Batch constraints:** cakes produced must respect batch sizes and minimum quantities.
- **Channel capacity:** number of cakes sold in each channel cannot exceed service limits.

- **Budget constraint:** total investment does not exceed \$4,000.
- **Non-negativity and integrality:** production and sales variables must be non-negative integers.

Stage 4: Logical Constraints

At Least Two Channels per Cake Type. Let $z_{ij} \in \{0, 1\}$ indicate whether cake type i is sold in channel j (Local, Supermarket, Online Local). **Formulate constraints** that guarantee each cake type i is sold in *at least two* distinct channels. Provide the necessary justification. Properly define any additional constants you introduce.

Supply-Threshold Activation (Supermarket). A supermarket does not accept very small quantities of cakes. Accordingly, for each cake of type i , the number of cakes supplied to the supermarket can either be 0 or above a minimum threshold of 15 units. Formulate the necessary constraints to satisfy this requirement.

Stage 5: Report and Deliverables

Each team must submit a comprehensive report including:

1. **Basic Formulation:** Formulate the ILP without the logical constraints. Clearly define the decision variables, the objective function, and all the constraints.
2. **Basic Solution:** Using Python, solve the ILP for different prices and generate the associated optimal production plans. Choose the best set of prices that you reach and the associated optimal production plan. Explain the logic you adopted to find good prices.
3. **Analysis of Outputs:** For the best set prices you found and the associated optimal production plan, show the detailed costs and revenues breakdown (For example by cake type, by channel, etc.). Illustrate your discussion with charts and figures.
4. **Full Formulation:** Update your ILP formulation to include the logical constraints.
5. **Full Solution:** Using Python, solve the updated ILP and generate a good set of prices and the associated optimal production plan.
6. **Comparison and Conclusion:** Compare the Basic Solution (without the logical constraints) to that after introducing the logical constraints.

Competition and Evaluation

This project forms the first phase of the **Sweet Market Competition**. All teams will compete using identical data files and the same initial capital. Your performance will be evaluated based on:

- Correct and consistent model formulation.
- Soundness of investment and pricing strategy.
- Quality of analysis and discussion.
- Final achieved profit.

A **profit leaderboard** will show profits from each team's submission. The teams achieving the highest profit will be recognized as the best-performing shops for this phase. Further phases will be announced soon.

Prepare your models carefully, test your assumptions, and develop strong insights—this preparation will be crucial for later competitive rounds.